Characteristics and classification of living organisms

Mark scheme and worked solutions for 20 questions: 10 multiple choice, 10 structured.

1 · Part 1

Core · Characteristics of living organisms · [1]

Answer: B

Explanation

Excretion is the removal of the waste products of metabolism and substances in excess of requirements. A describes respiration. C describes egestion: undigested food was never part of the body's metabolism, so removing it is not excretion. D describes nutrition.

2 · Part 1

Core · Concept and uses of classification systems · [1]

Answer: D

Explanation

A species is a group of organisms that can reproduce to produce fertile offspring. Only D shows that the offspring are fertile, because they go on to reproduce themselves. C shows that the frogs can produce offspring, but not that the offspring are fertile. A and B are shared features or behaviours, which can also be shared by different species.

3 · Part 1

Core · Concept and uses of classification systems · [1]

Answer: A

Explanation

In the binomial system the first part of the name is the genus and the second part is the species. *Pinus sylvestris* and *Pinus nigra* share the genus name *Pinus*. B pairs two trees that share only the second (species) part of the name, which does not put them in the same genus. C and D pair trees with different genus names.

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4 · Part 1 Core Concept and uses of classification systems [1]

Answer: C

Explanation

Start at step 1: the fish has no barbels, so go to step 3. At step 3 the body has stripes, so the fish is R. The forked tail fin is only used at step 2, which this fish never reaches, so P is wrong. Q needs barbels, and S needs spots.

5 · Part 1 Supplement Concept and uses of classification systems [1]

Answer: A

Explanation

Species that share a more recent common ancestor have more similar DNA base sequences, so they have fewer bases that differ. Species Y has the fewest differences from species W. X has the most differences, so it is the least closely related; V and Z are more distantly related than Y.

6 · Part 1 Supplement Concept and uses of classification systems [1]

Answer: D

Explanation

Classification systems aim to reflect evolutionary relationships, so organisms are grouped by how recently they shared a common ancestor, not just by similar appearance. Sharks and dolphins look similar because both are adapted to swimming, but they share only a distant ancestor. A is wrong because many animals that cannot interbreed (for example two different species of mammal) are in the same vertebrate group. B is wrong because sharks and dolphins share a habitat yet are in different groups. C is wrong because all DNA is made of the same four bases; it is the sequence of bases that differs.

7 · Part 1 Core Features of organisms [1]

Answer: B

Explanation

Arachnids have two body sections, four pairs of legs and no antennae. Crustaceans have two pairs of antennae and more than four pairs of legs. Insects have three body sections, three pairs of legs and one pair of antennae. Myriapods have many body segments with legs on each segment and one pair of antennae.

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8 · Part 1 Core Features of organisms [1]

Answer: C

Explanation

Amphibians have smooth, moist skin without scales and lay eggs without shells in water, and their young (larvae) live in water. Fish lay eggs in water and have gills, but they have scales. Reptiles have dry, scaly skin and lay eggs with shells on land. Mammals have fur or hair and do not lay eggs in water.

9 · Part 1 Supplement Features of organisms [1]

Answer: D

Explanation

Fungal cells have a nucleus and a cell wall, but no chlorophyll, so fungi cannot photosynthesise. A has no nucleus, so it describes a prokaryote. B has no cell wall, so it cannot be a fungus (it fits an animal cell). C has chlorophyll, which fits a plant cell, not a fungus.

10 · Part 1 Supplement Features of organisms [1]

Answer: B

Explanation

A virus is made of genetic material surrounded by a protein coat. Viruses are not cells, so they have no cell membrane, cytoplasm, cell wall or nucleus. A, C and D each include a cell structure that viruses do not have.

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11 · Part 1 Core Characteristics of living organisms [5]

Mark scheme

(a)(i) **[1]**

- sensitivity (1)

(a)(ii) **[1]**

- respiration (1)

> **REJECT** breathing

(b) **[2]**

- any two from: respiration; sensitivity; nutrition; excretion; reproduction; movement (1) (1)

> **IGNORE** growth

> **REJECT** breathing as an alternative to respiration

(c) **[1]**

- fresh mass includes water, and the water content can go up or down temporarily (e.g. after watering or when the plant wilts), so only an increase in dry mass shows a permanent increase (1)

> **ALLOW** fresh mass can increase without growth because the seedling takes up water

Worked solution

(a) Each statement is the definition of one of the characteristics of living organisms. Detecting and responding to changes is sensitivity. Chemical reactions in cells that break down nutrient molecules and release energy are respiration. Breathing is not the same as respiration: breathing moves air, but respiration is a set of chemical reactions in cells.

(b) The salt crystal gets bigger only because more salt from the solution is deposited on it. It has no cells, so it does not carry out any of the other life processes. The bean seedling respires, responds to light and gravity (sensitivity), takes in carbon dioxide, water and mineral ions (nutrition), removes waste such as oxygen and carbon dioxide (excretion), can reproduce and can move parts of itself, for example turning its leaves towards light. Any two of these earn the marks. Growth is excluded because the question asks for characteristics other than growth.

(c) A plant's fresh mass changes a lot with how much water it contains. Watering a seedling can raise its fresh mass in an hour, and it can fall again if the plant wilts, but that is not growth. Growth is a permanent increase in size and dry mass, so scientists remove the water (by drying) and measure the dry mass.

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12 · Part 1 Core Concept and uses of classification systems [6]

Mark scheme

(a) **[4]**

- first pair of statements uses one feature, with opposite statements, to divide the four fruits into two groups (1)
- second pair of statements correct, with opposite statements about one feature (1)
- third pair of statements correct, with opposite statements about one feature (1)
- each of E, F, G and H appears once at the end of a branch and is correctly identified (1)
- example: 1 fruit is fleshy, go to 2 / fruit is not fleshy, G; 2 fruit has many seeds, F / fruit has one seed, go to 3; 3 skin is hairy, E / skin is smooth, H

ALLOW any order of features that works, e.g. starting with surface of skin or number of seeds

ALLOW keys written as numbered statements, a table or a branching outline

REJECT a pair of statements that are not about the same feature, e.g. "fruit is fleshy / fruit has many seeds" (no credit for that pair)

(b) **[2]**

- colour varies between fruits of the same type / colour changes as the fruit ripens (1)
- so fruits of the same type could be identified as different fruits / the key would give a wrong identification / a key needs features that are the same in all fruits of one type (1)

Worked solution

(a) Four fruits need three pairs of statements, because each pair splits a group into two. Pick a feature that separates one fruit from the other three: G is the only fruit that is not fleshy. So step 1 is "fruit is fleshy, go to 2" or "fruit is not fleshy, G". That leaves E, F and H, which are all fleshy. F is the only one with many seeds, so step 2 is "many seeds, F" or "one seed, go to 3". That leaves E and H; E has a hairy skin and H has a smooth skin, so step 3 separates them. Each pair of statements must be about one feature and must be opposites, so that every fruit fits exactly one statement. Other orders work too, as long as every fruit ends up on its own.

(b) A good key uses features that are the same in every individual of a type. The question tells you that fruits of one type can be green, yellow or red, for example as they ripen. If the key used colour, a green fruit and a red fruit of the same type would end up at different places in the key and be named as different fruits.

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13 · Part 1 Core Concept and uses of classification systems [6]

Mark scheme

(a) **[1]**

- *Canis* (1)

ALLOW Canis written without italics

REJECT *Canis latrans* (this is the full scientific name, not the genus)

(b) **[2]**

- Arctic fox (1)
- both are in the same genus / both have the genus name *Vulpes* / the first part of their scientific names is the same (1)
- the second mark depends on the first

(c) **[2]**

- a group of organisms that can reproduce / interbreed (1)
- to produce fertile offspring (1)

(d) **[1]**

- the scientific name is internationally agreed / the same in every country and language, so scientists everywhere know exactly which organism is meant / avoids confusion (1)

Worked solution

(a) In the binomial system the scientific name has two parts. The first part, written with a capital letter, is the genus. The second part is the species. The coyote is *Canis latrans*, so its genus is *Canis*.

(b) Organisms in the same genus share more features and are more closely related than organisms in different genera. The red fox is *Vulpes vulpes*. The Arctic fox, *Vulpes lagopus*, is the only other mammal in the table in the genus *Vulpes*, so it is the most closely related.

(c) A species is a group of organisms that can reproduce to produce fertile offspring. Both ideas are needed: "can reproduce" alone is not enough, because some different species can mate but their offspring are infertile.

(d) Common names change from country to country and language to language, and one common name can even be used for different animals. The binomial system is internationally agreed, so *Vulpes vulpes* means the same animal to every scientist in the world.

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14 · Part 1 Supplement Concept and uses of classification systems [6]

Mark scheme

(a) **[1]**

- 7 (1)

(b) **[3]**

- L (1)
- the base sequence of L has the fewest differences from K / L has only 2 bases that differ from K / the DNA of L is the most similar to K (1)
- so L shares the most recent common ancestor with K (1)
- the second and third marks depend on identifying L

(c) **[2]**

- DNA (and its base sequence) is passed on / inherited from ancestors (1)
- organisms that share a more recent common ancestor have more similar base sequences, while organisms that share only a distant ancestor have more differences (1)
- organisms that are not closely related can have similar external features, so base sequences give more reliable evidence of how closely related organisms are (1)
- max 2

Worked solution

(a) Compare species K and species M one base at a time, group by group. Group 1 (ATGCC and ATCGC): the 3rd and 4th bases differ, so 2. Group 2 (TAGGA and TTGGA): the 2nd base differs, so 1. Group 3 (TCCTA and GCCAA): the 1st and 4th bases differ, so 2. Group 4 (AGCTT and AACTA): the 2nd and 5th bases differ, so 2. Total = 2 + 1 + 2 + 2 = 7. Check by counting the matches instead: 13 of the 20 bases match, and 20 - 13 = 7.

(b) The fewer bases that differ, the more similar the DNA. L differs from K at only 2 positions, N at 4 and M at 7. Similar DNA shows that two species have a recent common ancestor, so L is the most closely related to K.

(c) Organisms inherit their DNA from their ancestors. Over time, the base sequences in separate groups change, so groups that split from a common ancestor long ago have more differences, and groups that split recently have fewer. Comparing base sequences therefore shows how closely related organisms are, which is what classification systems aim to show. Appearance can be misleading, because unrelated organisms can look alike.

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15 · Part 1 Core Concept and uses of classification systems [5]

Mark scheme

(a) **[3]**

- leaf 1: Z (1)
- leaf 2: W (1)
- leaf 3: X (1)

(b) **[1]**

- leaves of the same type of tree vary in size / young leaves are smaller than older leaves, but the shape of the leaf (its length compared with its width) stays about the same (1)

(c) **[1]**

- by the features that they share (1)

ALLOW by their similarities

Worked solution

(a) Leaf 1 is not divided into leaflets, so step 1 sends you to step 3. Its edge is toothed, so go to step 4. It is 8 cm long and 5 cm wide; twice its width is 10 cm, and 8 cm is less than 10 cm, so it is less than twice as long as it is wide. Leaf 1 is Z. Leaf 2 is divided into leaflets, so go to step 2; its leaflets are in two rows along a central stalk, so it is W. Leaf 3 is not divided into leaflets (go to step 3) and has a smooth edge, so it is X.

(b) Leaves on one tree come in many sizes, because they grow and because some get more light or water than others. A key that used actual length would give different answers for a small leaf and a large leaf from the same tree. The shape (length compared with width) stays much the same, so it is a more reliable feature.

(c) Organisms are classified into groups by the features that they share: organisms with many features in common are placed in the same group. A dichotomous key uses features in a similar way, separating organisms by features that can be seen.

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16 · Part 1 Core Features of organisms [4]

Mark scheme

(a) **[2]**

- any two from: multicellular; cells have no cell walls; no chlorophyll / no chloroplasts / cannot photosynthesise; feed on organic substances made by other living organisms (1) (1)

ALLOW cells have nuclei; usually have nervous coordination; able to move from place to place

(b) **[2]**

- it has chlorophyll / chloroplasts, so it photosynthesises / makes its own food (1)
- its cells have cell walls (made of cellulose) (1)

IGNORE multicellular (animals are also multicellular)

Worked solution

(a) The features used to place an organism in the animal kingdom are: it is multicellular; its cells have no cell walls; it has no chlorophyll, so it cannot photosynthesise; and it feeds on organic substances made by other living things.

(b) Organisms are classified by their features, not by one unusual behaviour. The Venus flytrap's leaves are green because they contain chlorophyll, so it photosynthesises and makes its own food; the insects it digests are only an extra source of mineral nutrients. Its cells also have cellulose cell walls. Both of these are plant features that animals never have, so the Venus flytrap is a plant. Being multicellular does not help, because plants and animals are both multicellular.

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17 · Part 1 Core Features of organisms [6]

Mark scheme

(a) **[4]**

- J: arachnid (1)
- K: insect (1)
- L: myriapod (1)
- M: crustacean (1)

(b) **[2]**

- crustaceans and myriapods also have antennae (1)
- insects are identified by other features as well, such as a body in three sections / three pairs of legs / one pair of antennae / wings (1)

ALLOW crustaceans have two pairs of antennae for the first mark

Worked solution

(a) J has two body sections, four pairs of legs and no antennae, so it is an arachnid. K has a head, thorax and abdomen, three pairs of legs, one pair of antennae and wings, so it is an insect. L has many similar segments with legs on each one, so it is a myriapod. M has two pairs of antennae and more than four pairs of legs, so it is a crustacean. The exoskeleton does not help with M, because all arthropods have one.

(b) Having antennae does not make an arthropod an insect. Myriapods have one pair of antennae and crustaceans have two pairs; only arachnids have none. To place an arthropod in the insects you need features that only insects have together: three body sections, three pairs of legs and one pair of antennae, and usually wings.

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18 · Part 1 Core Features of organisms [6]

Mark scheme

(a) **[3]**

- F: reptile (1)
- G: bird (1)
- H: fish (1)

(b) **[2]**

- any two from: bats have fur / hair, not feathers; bats have external ears / pinnae; female bats have mammary glands / feed their young on milk; bats have teeth, not a beak; bats give birth to live young (1) (1)

IGNORE wings / can fly

(c) **[1]**

- has feathers / has a beak / lays eggs with hard shells (1)

REJECT has wings / has flippers

Worked solution

(a) F has dry, scaly skin and lays soft-shelled eggs on land, which are reptile features. G has feathers and a beak and lays hard-shelled eggs, which are bird features. H has scales and fins and breathes with gills, which are fish features.

(b) Animals are classified by the main features of their group, not by whether they can fly. Bats have fur, not feathers. They also have external ears, and females have mammary glands and feed their young on milk. These are mammal features, so bats are mammals. Wings are not a good feature to use, because both birds and bats have them.

(c) Penguins cannot fly, but they still have the main features of birds: a body covered in feathers, a beak, and eggs with hard shells. Any one of these shows that a penguin is a bird.

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19 · Part 1 Supplement Features of organisms [6]

Mark scheme

(a) **[4]**

- P: prokaryote (1)
- Q: fungus (1)
- R: protoctist (1)
- S: plant (1)

(b) **[2]**

- R is made of one cell / is unicellular, but plants are multicellular (1)
- R has no cell wall, but plant cells have cell walls made of cellulose (1)

Worked solution

(a) P has no nucleus, and its DNA is circular with plasmids, so it is a prokaryote. Q is made of hyphae, has cell walls made of chitin and no chlorophyll, and feeds on dead material, so it is a fungus. R is a single cell with a nucleus. It cannot be a prokaryote (it has a nucleus), an animal (it has chlorophyll), a plant (it is single-celled with no cell wall) or a fungus (no cell wall and it has chlorophyll), so it is a protoctist. S is multicellular, with a nucleus, cellulose cell walls and chlorophyll, so it is a plant.

(b) Having chlorophyll is not enough to make an organism a plant. Plants are multicellular and their cells have cellulose cell walls. R is a single cell and has no cell wall, so it does not have the main features of the plant kingdom. Some protoctists have chlorophyll and photosynthesise, like plants.

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20 · Part 1 Supplement Features of organisms [6]

Mark scheme

(a) **[2]**

- any two from: leaves are called fronds; does not produce flowers; reproduces by spores / does not produce seeds (1) (1)

(b) **[3]**

- T: monocotyledon (1)
- U: dicotyledon (1)
- X: dicotyledon (1)

(c) **[1]**

- one (1)

Worked solution

(a) Ferns are plants with leaves called fronds. They do not make flowers or seeds; instead they reproduce by spores. Flowering plants produce flowers, and their seeds develop inside the flower.

(b) Monocotyledons usually have leaves with parallel veins and flower parts in threes or multiples of three. Dicotyledons usually have a branching network of veins and flower parts in fours or fives, or multiples of these. T has parallel veins and 6 petals (a multiple of three), so it is a monocotyledon. U has net veins and 5 petals, and X has net veins and 4 petals, so both are dicotyledons.

(c) The name tells you the number of cotyledons: a monocotyledon has one cotyledon in its seed and a dicotyledon has two. T is a monocotyledon, so its seed has one cotyledon.

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